

ShareSpace®

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COURSE:

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1 PROJECT DESCRIPTION

1.1 GENERAL DESCRIPTION

ShareSpace® is a community-driven marketplace that supports the growing sharing economy by allowing users to lend and rent items that are not in constant use. The system provides customization and management of user profiles to enhance trust between users and create a personalized experience. Also, it provides financial calculations to ensure a reliable exchange of goods and a user-friendly way to create and manage [listed] items. Furthermore, the system coordinates the logistical aspects of booking and ensures reliable usage of the software.

1.2 TECHNICAL ASPECTS

ShareSpace® follows a layered software architecture to ensure maintainability, scalability, and clear separation of responsibilities. The System is based on a client-side SQLite database for simplicity. Data access is handled through dedicated DAO (Direct Access Object) components, while business logic is managed within the service layer.

The graphical user interface and user experience is created using a modern design implemented through Figma and JavaFX. UML diagrams model the database structure and system layers, providing a structured overview of the application architecture and components.

2 MILESTONE 1 - TECHNICAL AND VISUAL PROJECT FOUNDATION

2.1 TECH STACK

2.1.1 OpenJDK25

The project is developed using OpenJDK version 25. We chose this version because it is the latest long-term support version of Java.

2.1.2 SQLite

The Database system is SQLite, which is a lightweight, file-based relational database that does not require a separate server. It runs directly within the client application, which means all data is stored and managed locally.

2.1.3 Java Database Connectivity

Database access is handled through JDBC, because it allows Java programs to connect to a database, run queries, retrieve and manipulate data.

2.1.4 Figma

The UI/UX prototype was designed using Figma. We chose Figma because it has a fast learning curve and changes are visible directly, there is no compiling time. Also, it uses cloud storage, so group work is simplified.

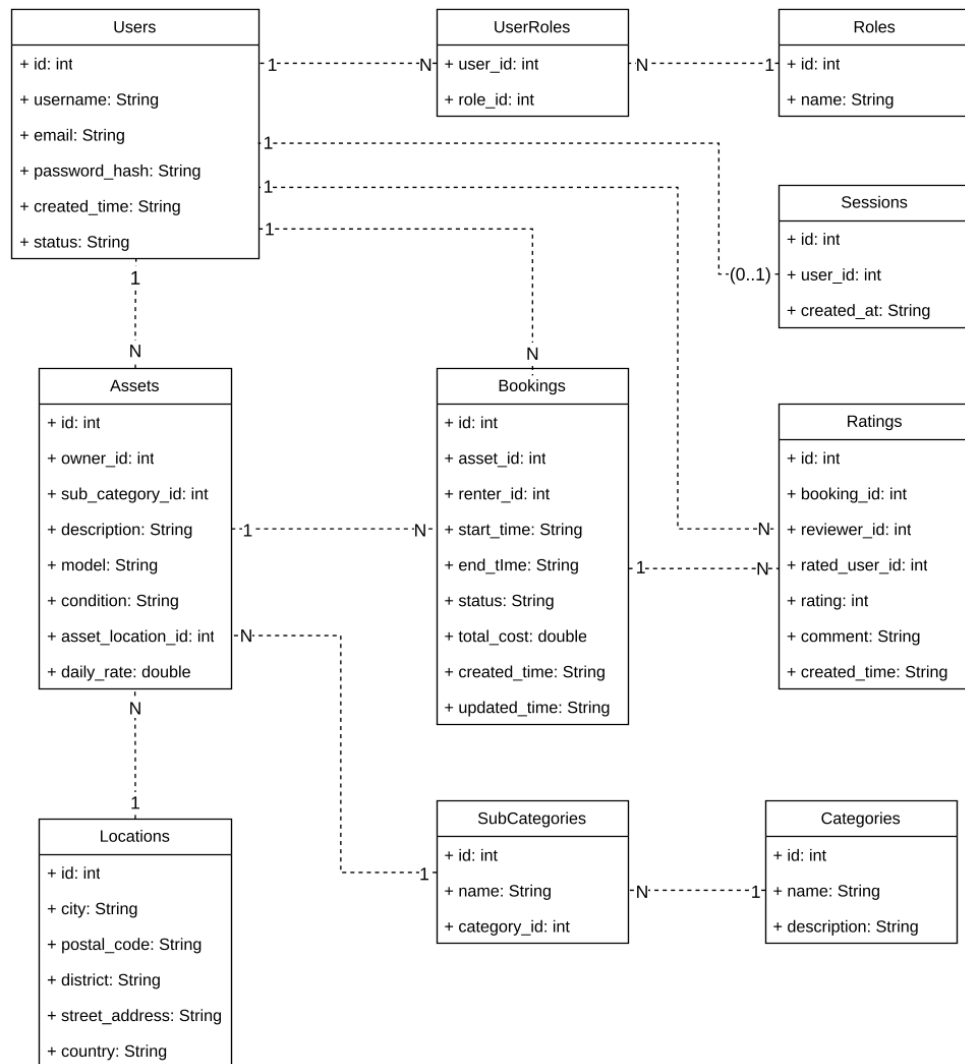
2.1.5 JavaFX

The user interface is [probably] implemented using JavaFX because it is a newer framework which is actively maintained and has great performance. It is seamlessly integrated with modern Java versions (OpenJDK 25) and it has cross-platform compatibility.

2.2 UML DOCUMENTATION

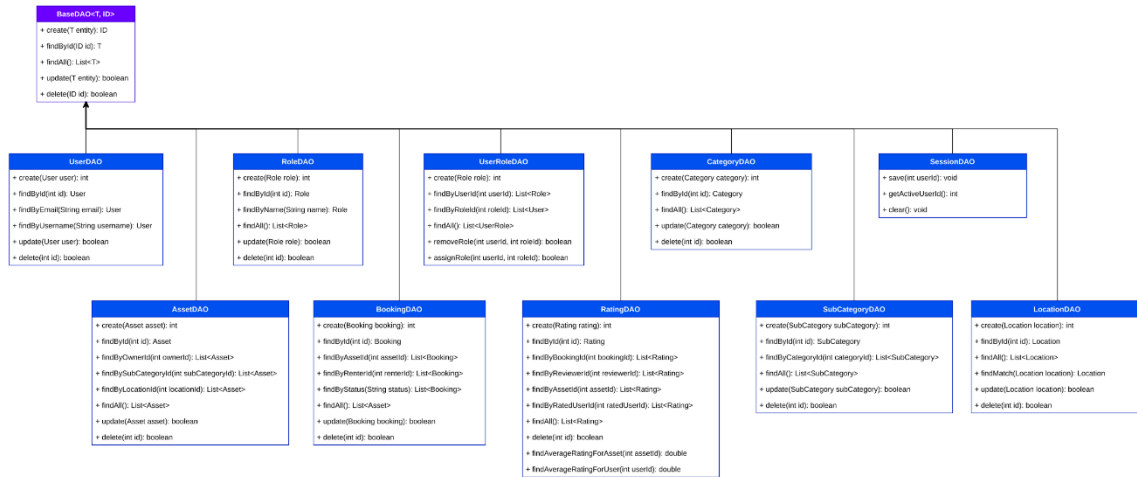
2.2.1 Database

ShareSpace Database Schema UML



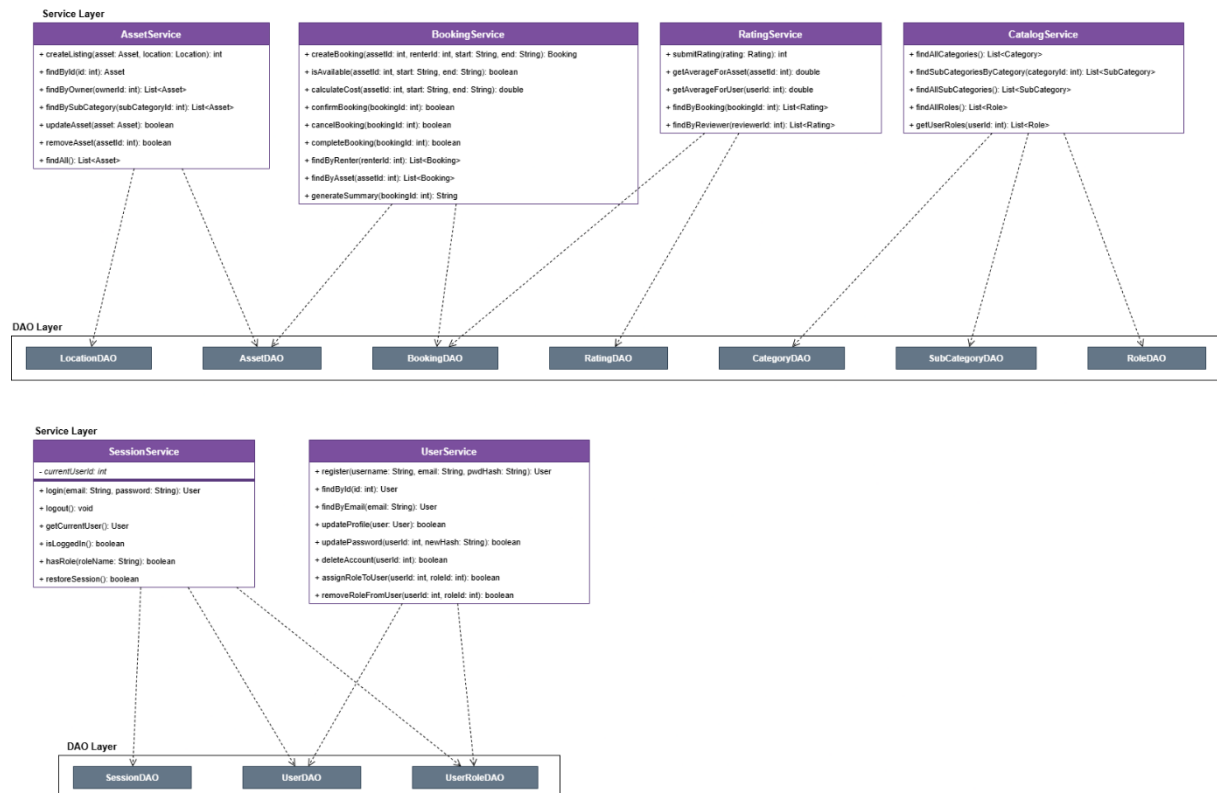
2.2.2 Dao

DAO Architecture UML



2.2.3 Services

ShareSpace – Service Layer UML



2.3 UI/UX CONCEPT

2.3.1 Homepage

ShareSpace®

[SHARE](#)

[ABOUT](#)

[LOGIN](#)

Rechtliches | Impressum

ShareSpace®



2.3.2 Login

[← Back](#)

[Create an account](#)

Log in

Log in

Email address

Password

 Hide

Log in

OR

 Continue with Google

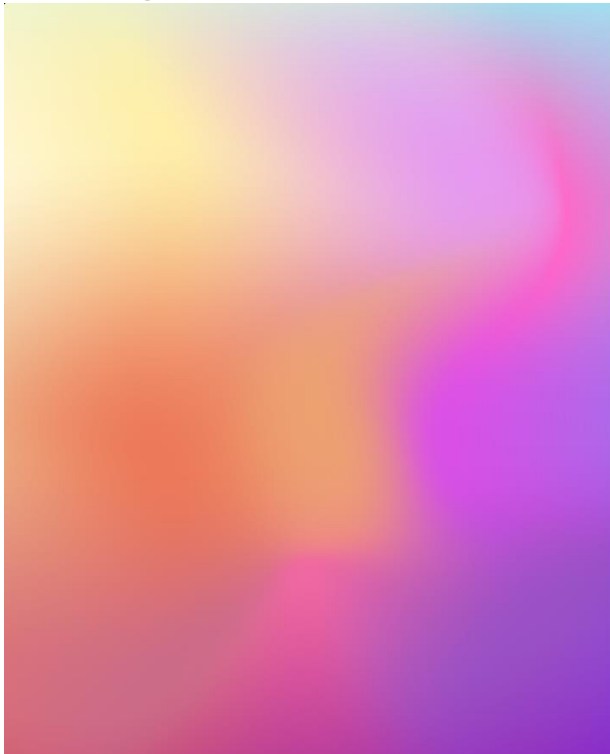
 Continue with Facebook

 Sign up with email

[Can't log in?](#)

Secure Login with reCAPTCHA subject to
Google [Terms](#) & [Privacy](#)

2.3.3 Signup



Create an account

Already have an account? [Log in](#)

User name

Email address

Password

 Hide

Use 8 or more characters with a mix of letters, numbers & symbols

By creating an account, you agree to our
[Terms of use](#) and [Privacy Policy](#)

☒ I'm not a robot



Create an account

Already have an account? [Log in](#)

3 MILESTONE 2

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